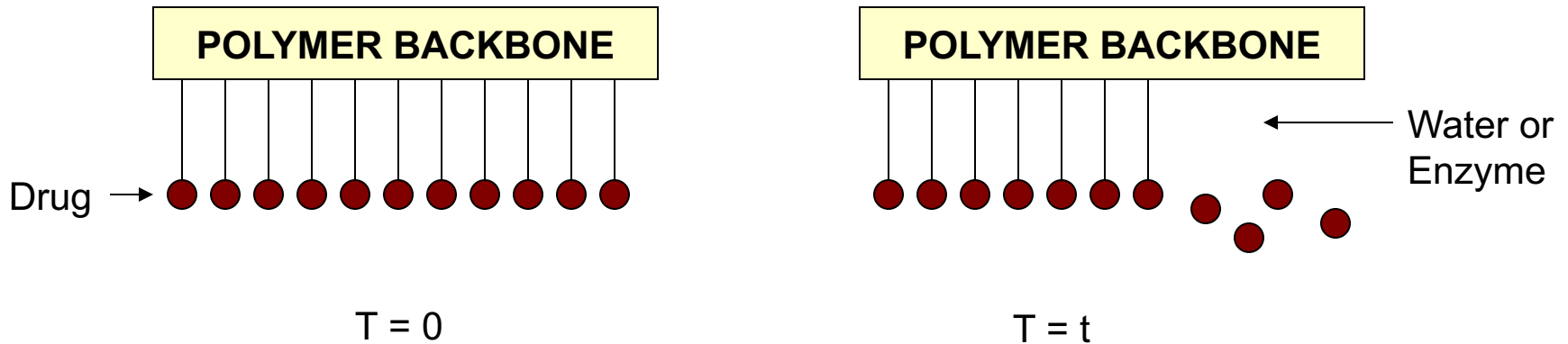
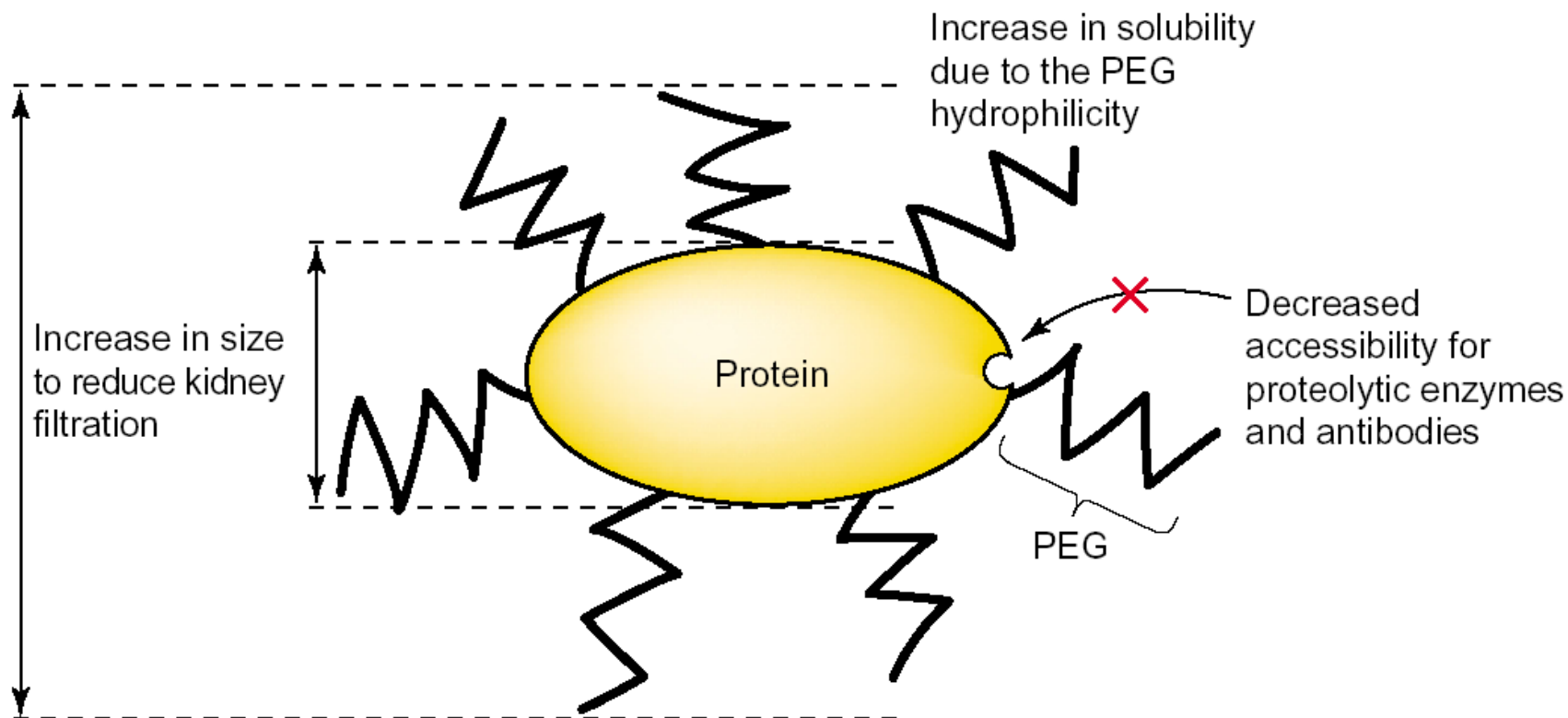


Polymer-drug conjugates: Small molecule



- Primary advantage
 - Very high drug loading
- Primary disadvantage
 - New Chemical entity

Protein-Polymer Conjugates for large molecules



Drug Discovery Today

Protein-polymer therapeutics are **nano-sized** drugs with many advantages

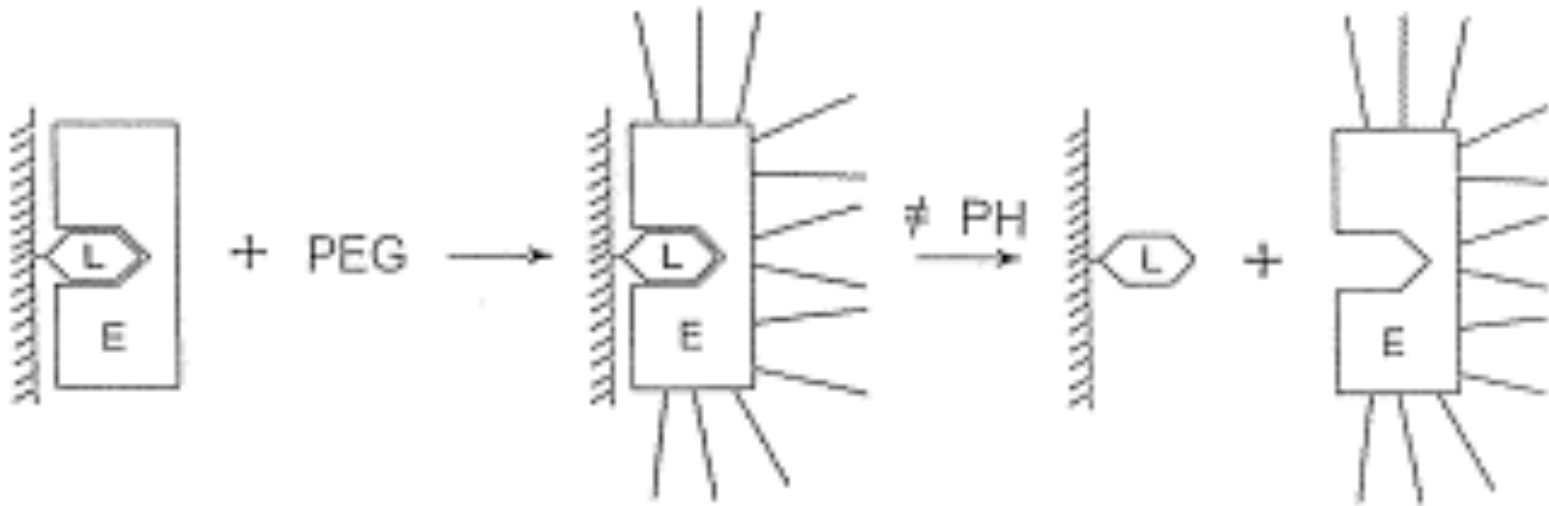
Advantages of polymer conjugates

- Modify chemical properties – Enhance Solubility
- Protection of labile drugs from biochemical degradation - Stability
- Increased half-life (body residence time) - Stability
- Reduction of immunogenicity and toxicity - Safety
- Improved targeting (to specific organs and specific cells) – Specificity and bioavailability, Controlled delivery
- Drug conjugation via degradable bonds – Controlled release
- Drug conjugation to stimuli responsive polymers – Delivery under specific conditions – Controlled release

Limitations of polymer-drug conjugates

- **Bio-molecule (drug) can loose activity during conjugation process** – careful choice of chemistry and reaction sites.
- **Purification** of un-reacted polymer, bio-molecules and any reaction products is necessary before in-vivo application
- **Sterilization** process now involves polymer stability
- Can be viewed by regulatory authorities as a **“new” drug entity** (novel chemical composition) even though the drug molecule itself is well used: could take years and years along with high cost

Active site protection

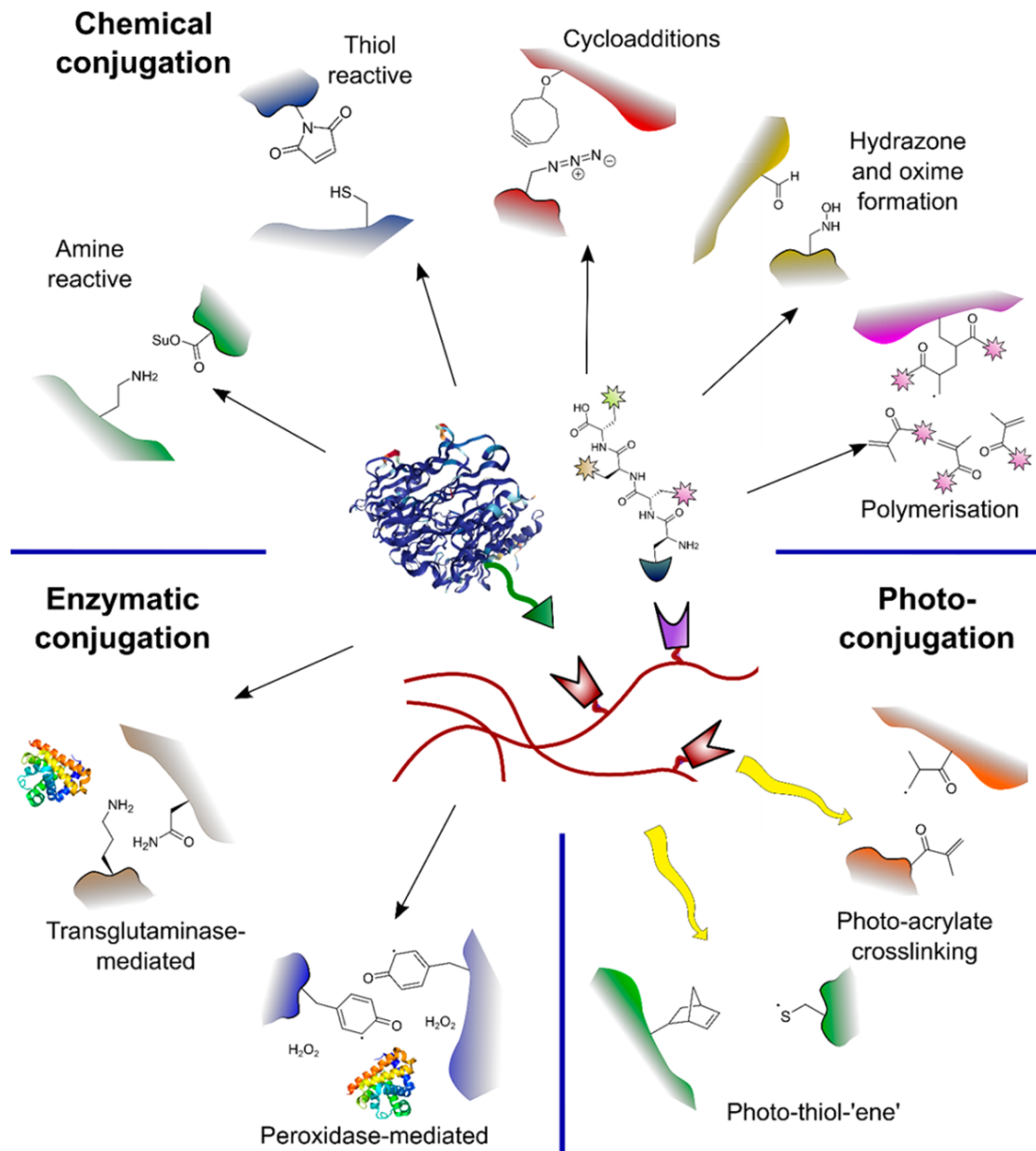


- Fig. A ligand, such as a substrate or inhibitor (L), linked to an insoluble matrix gives reversible binding with the enzyme or protein and prevents the PEGylation around the binding site. The PEGylated macromolecule is released from the ligand by any condition that reverses the binding such as pH, salts, solvents.

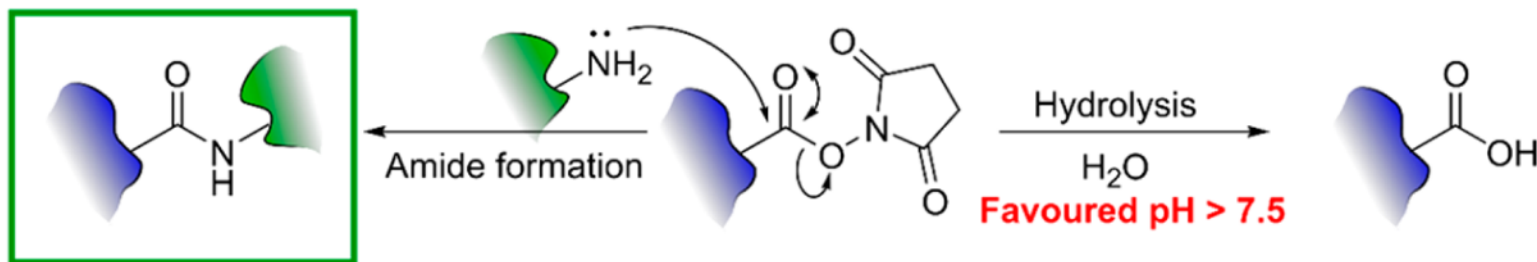
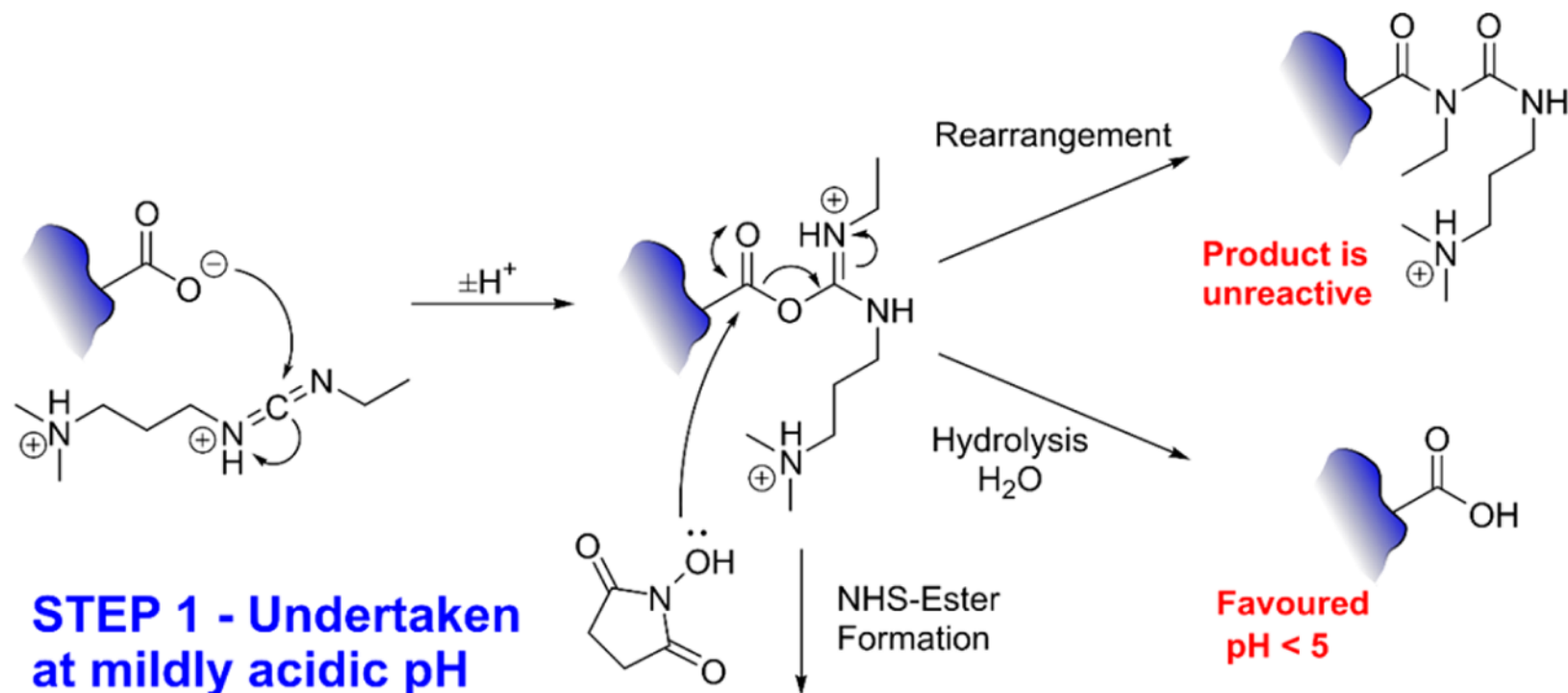
Polymers in drug conjugation

- Mostly synthetic polymers are used for conjugation owing to their controllable structure and properties
 - Exceptions, natural polysaccharides e.g. Dextrans
- Usually water soluble (hydrophilic), highly biocompatible polymers containing reactive functional groups are used
- Most commonly used polymers
 - Polyethylene glycol (PEG) and derivatives
 - N-(2-hydroxypropyl) methyl acrylamide (HPMA)
 - Dextrans and other polysaccharides
 - Poly(amino acids)
 - Stimuli-sensitive polymers etc.

Biomolecule-Biomaterial Conjugation



Biomolecule-Biomaterial Conjugation: EDC coupling



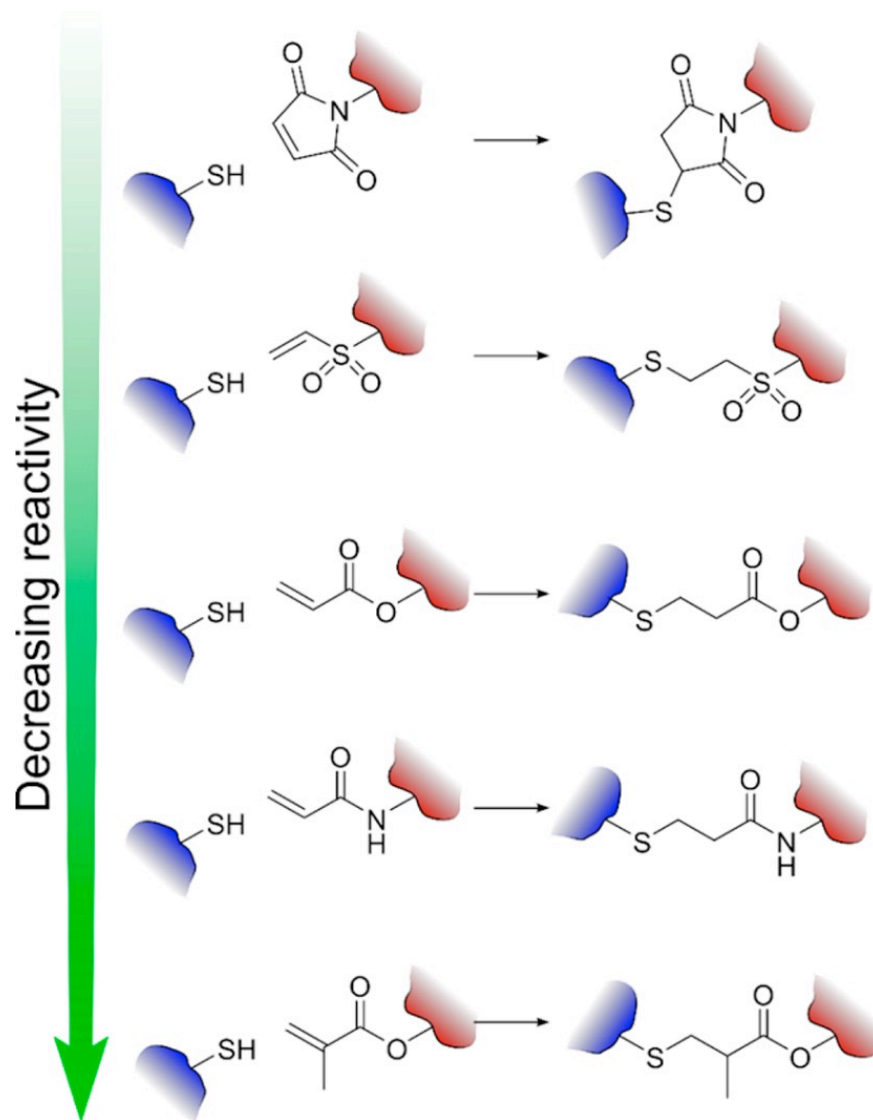
STEP 2 - Undertaken at neutral pH

Biomolecule-Biomaterial Conjugation: Thiols – Michael Addition

Less than 2% proteins have surface thiols and hence reactions are highly site specific

But at the same time they are usually involved in active site or control protein function so can be counterproductive

Within the most widely utilized acceptors in the biomaterial field, a trend of reactivity can be generalized as maleimides > vinyl sulfones > acrylates > acrylamides > methacrylates.



Functional groups on the drug:

Small molecules

- Small molecule drugs
 - Normally conjugation is performed with nucleophilic residues on the drug molecule
 - Hydroxyl (OH), Amine (NH₂) or Carboxyl (COOH) groups
 - The reactive group must not be of critical relevance for drug activity

Functional groups on the drug:

Proteins and Peptides

- Amines (Lysine, Arginine, 5' end amine, Histidine): Most widely used residue for conjugation. 10% of overall amino acids is lysine and only few are involved in the active site.
- Sulfhydryls (Cysteine): Highly reactive, but present only in few cases, and also responsible for dimer, trimerizations.
- Carboxyls (Aspartic and Glutamic acids):
- Sugar moieties in glycoproteins or polysaccharides (carrying OH, amine or aldehydes) are also used for conjugations

Reactive groups in polymers

- Polymers used in drug conjugation:
 - PEG, HPMA, Dextrans, Poly(L-lysine), poly(amino acids), polyvinyl pyrrolidone (PVP), etc.
- Primary reactive groups present in polymers:
 - OH, NH₂, COOH that are already present or derivatized
- Three distinct strategies for conjugation:
 - Reaction of the drug with functional groups present on the polymer chain
 - Reaction of polymer with an intermediate followed by drug attachment
 - Reaction of drug with an intermediate followed by polymer attachment

Example: Pre-derivatization of Dextrans

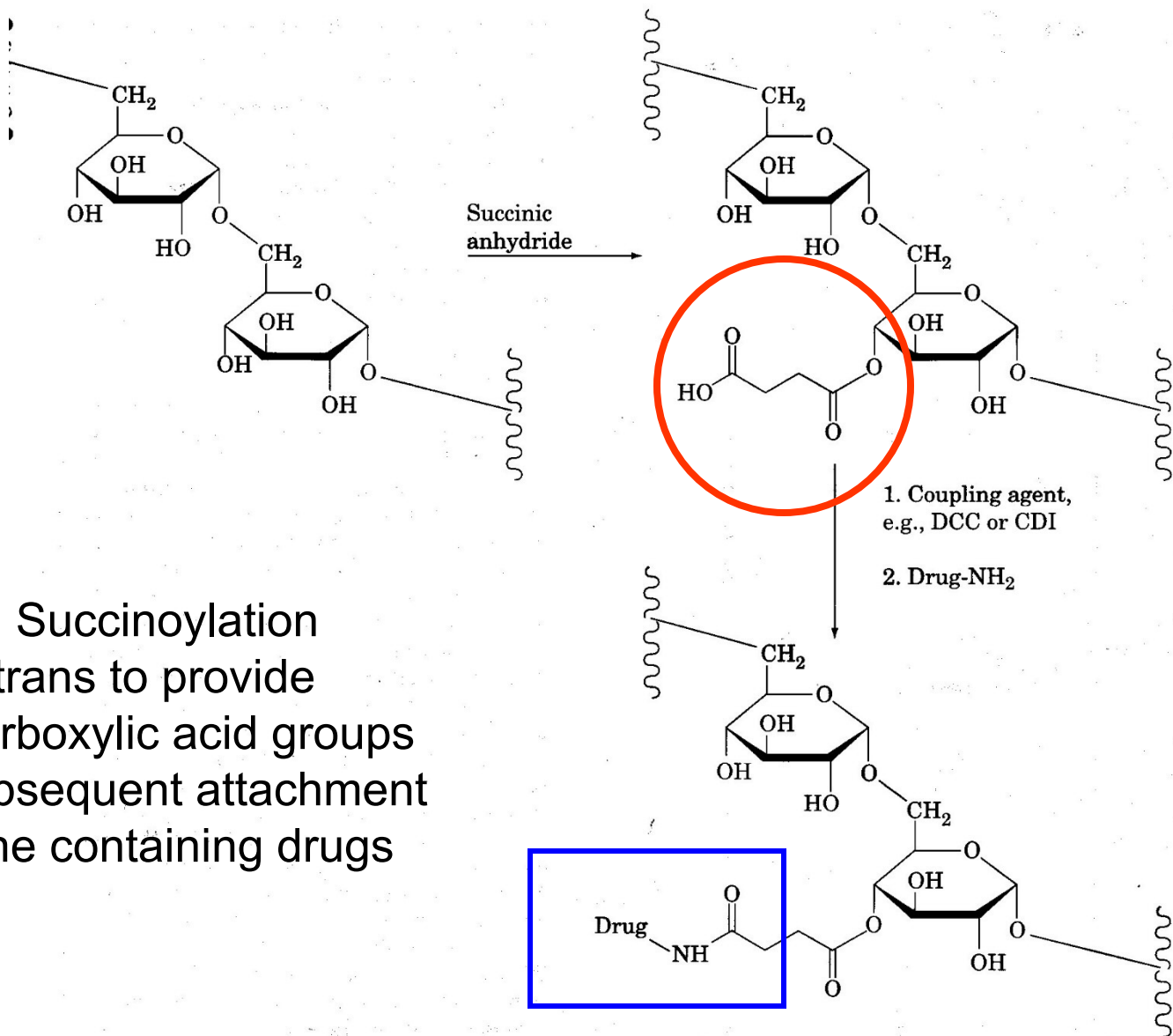
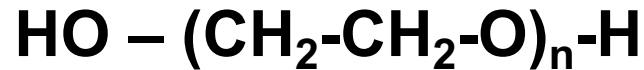


Figure: Succinylation of Dextrans to provide free carboxylic acid groups For subsequent attachment of amine containing drugs

Poly(ethylene glycol) [PEG]



- Very hydrophilic, highly biocompatible polymer
 - The backbone of the polymer is also highly flexible
 - Both these properties contribute to **“molecular windshield wiper”**
- Soluble in both aqueous and organic solvents
 - Well developed conjugation chemistries
- Non-toxic, non-immunogenic, can be produced under GMP (good manufacturing practice) conditions and FDA approved

Branched PEGylation

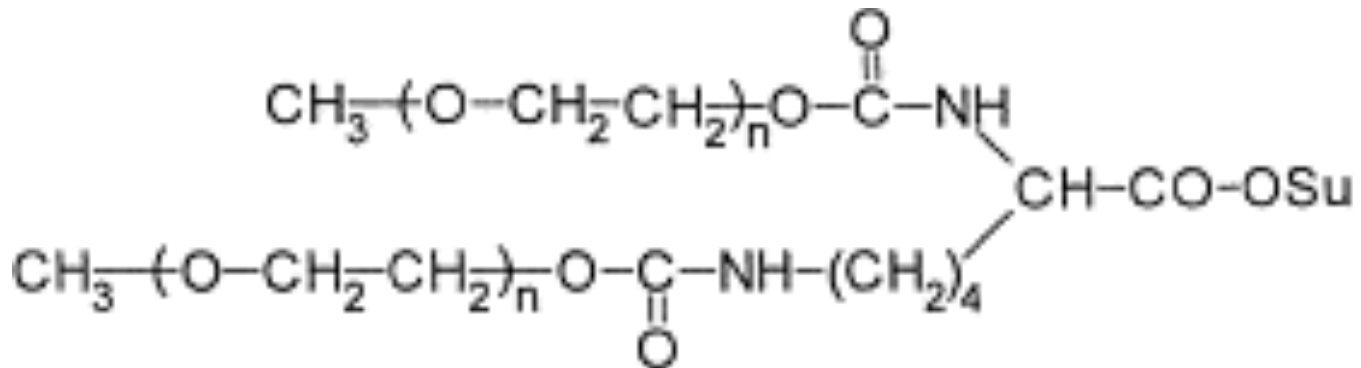


Fig. **Branched PEG (PEG2)**: Lysine is the amino acid spacer between two PEG molecules and the activated carboxylate. This PEG offers the advantages of higher protection of protein surface for single point of conjugation, as compared to the linear form. Generally, it conveys to its products higher retention in blood, lower immunogenicity and decreased enzyme inactivation during the conjugation.

Branched PEGylation

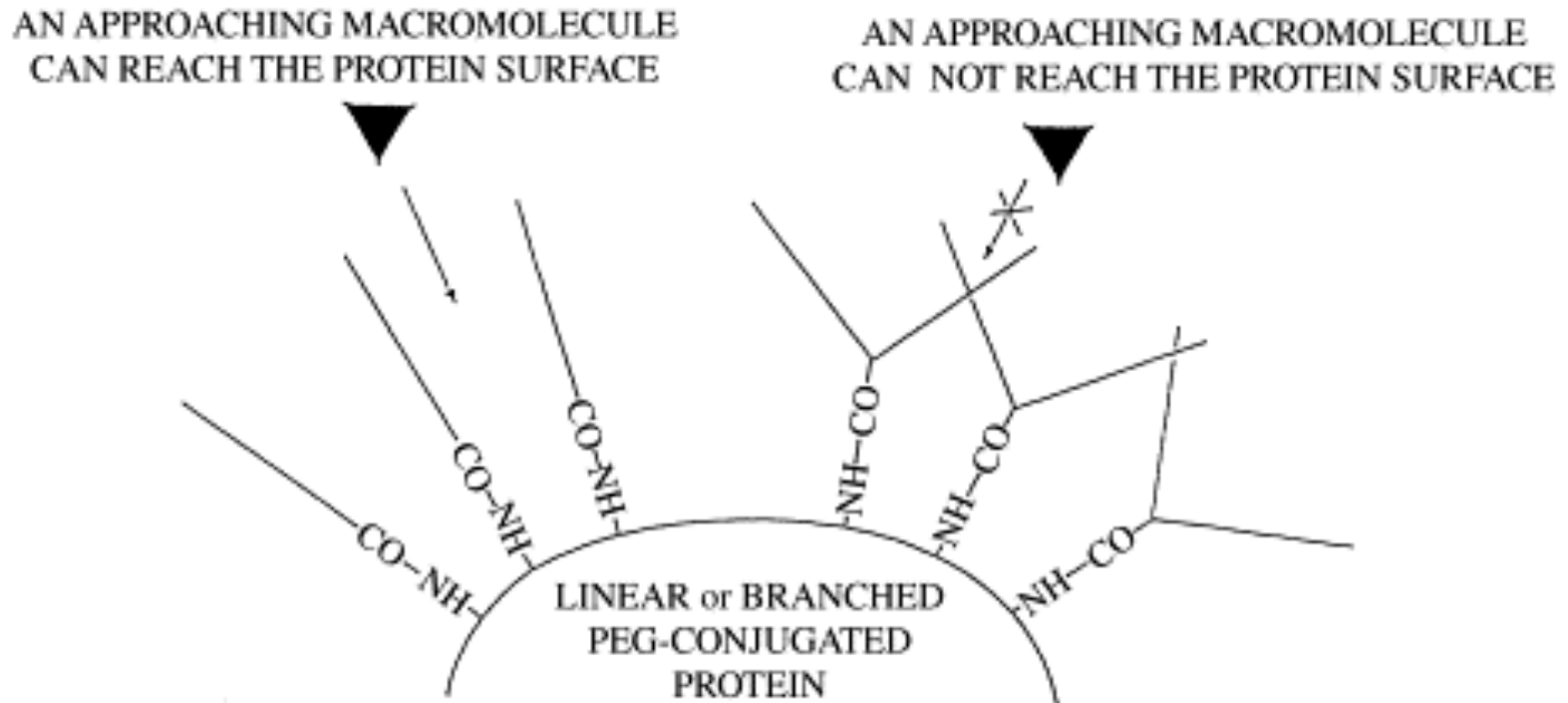
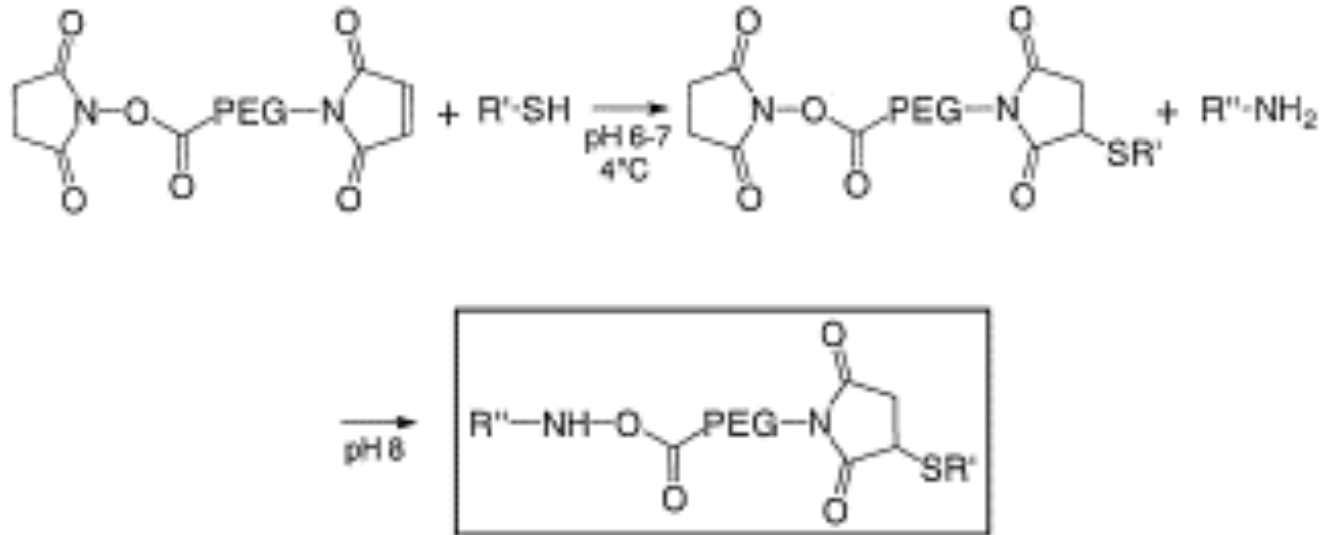


Fig. The "umbrella-like" structure of branched PEG explains the higher capacity in rejecting approaching molecules or cells as compared to linear PEG of the same size.

Linking multiple molecules



- Fig. -OSu- -maleimide-PEG: This is a typical cross-linking reagent that may reacts in a sequential order with different groups in protein, taking advantage of different pH and temperatures. With this strategy two different macromolecules may be stoichiometrically linked together.

PEGylation: solubility and clearance

- Water solubility of Taxol (a widely used chemotherapy drug) with or without PEG attachment

Mw of PEG	Solubility (mg/ml)
No PEG	~ 0
5,000	660
20,000	200
40,000	125

- Renal clearance and enzymatic degradation of proteins can be decreased thereby **increasing serum residence time** (SOD: Super-oxide dismutase)

Protein ^c	Half-life
SOD	0.08 ^a
SOD-PEG 1900 Da	1.5 ^a
SOD-PEG 5000 Da	11.0 ^a
SOD-PEG 72000 Da	36.0 ^b

Pharmacokinetics of PEGylated drugs



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Pharmacokinetic and biodistribution properties of poly(ethylene glycol)–protein conjugates

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Table 1

Physico–chemical and pharmacokinetic properties of PEGs: correlation among molecular weight, molecular size, protein molecular weight equivalence, permanence in the circulation after intravenous administration

PEG molecular weight (kDa)	PEG molecular size (nm) ^a	Protein molecular weight equivalence (gel permeation) ^b	Half time in blood (min) ^a
PEG 5 kDa	2	Cytochrome <i>c</i> (13 kDa)	18
PEG 10 kDa	–	Superoxide dismutase (32 kDa)	–
PEG 12 kDa	5	–	–
PEG 20 kDa	7	–	180
PEG 32 kDa	8	IgG (150 kDa)	–
PEG 40 kDa	10	–	–
PEG 48 kDa	11	–	840
PEG 73 kDa	–	Thyroglobulin (620 kDa)	–
PEG 180 kDa	24	–	1200
PEG 220 kDa	27	–	–

^a The data were obtained from the paper of Nakaoka et al. [16].

^b The data were obtained from the paper of Sherman et al. [13].

Properties of PEGylated drugs: Low immunogenicity

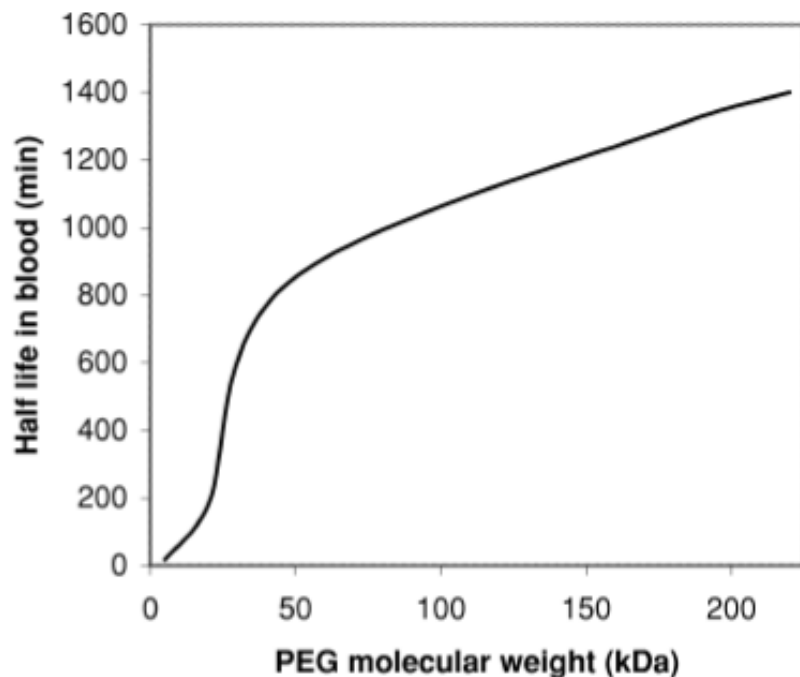


Fig. 1. Correlation plot between PEG molecular weight and half life in blood after intravenous administration. The data were obtained by the papers of Yamoka et al. [14] and Nakaoka et al. [16].

Table 2

Role of PEG shape on uricase immunogenicity^a

	Immunogenicity (%) ^b	
	IgG	IgM
Native uricase	100	100
Linear 5000 PEG uricase	0.2	0.3
Branched 10000 PEG uricase	0.2	0

^a The results were obtained from the paper of Caliceti et al. [45].

^b The immunogenicity study was carried out by mouse immunization with equimolar amount of PEG-enzyme or native enzyme. Uricase was modified with PEG to yield the same degree of derivatized amino groups. The protein immunogenicity of the various uricase forms was calculated as maximal serum dilution to reach the same enzyme-linked immunosorbent assay (ELISA) response and expressed as percentage with respect of the value obtained with the native enzyme.

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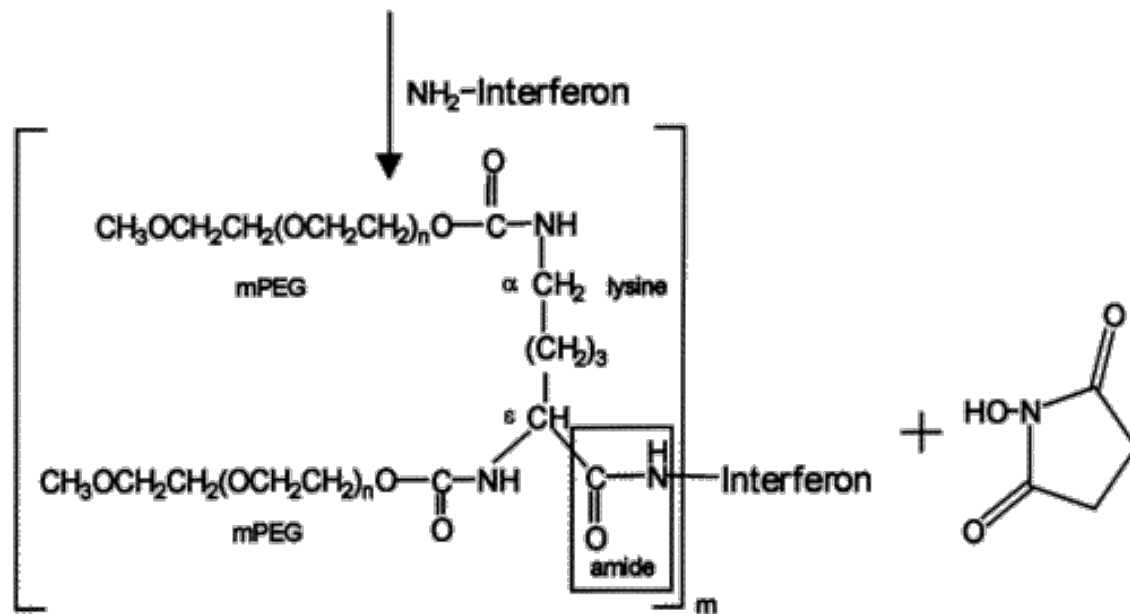
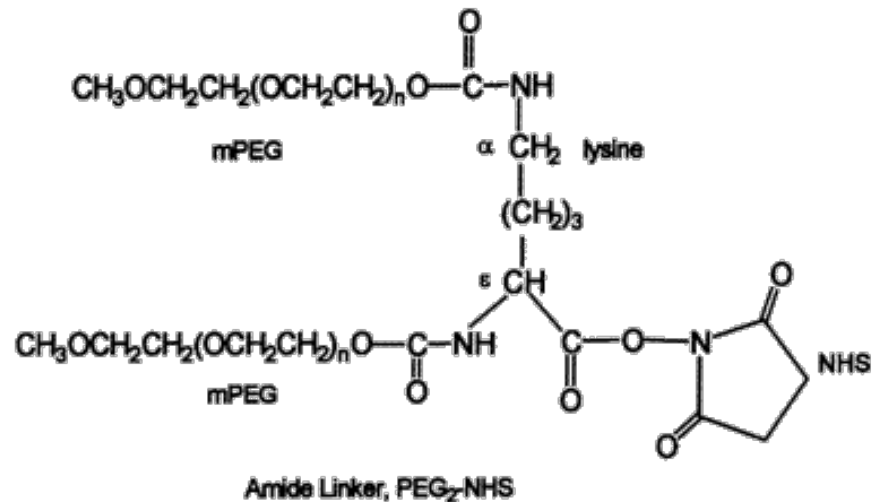
PEGylation: Specific Example

Interferon alpha -2a (IFN α -2a), Hoffmann-La Roche

IFN alpha is a cytokine with potent anti-viral and anti-tumor activities

- **Interferon alpha:**
 - **Terminal half-life** of interferon: 4 to 8 h, following intravenous (i.v.) or subcutaneous (s.c.) administration.
 - Little or no interferon is detected at 24 h
 - Necessitating frequent dosing of interferon. Certain oncological indications require daily dosing, while viral indications may require three times weekly dosing
 - Interferon treatment for these diseases: periods of several months to a year or longer

PEGylation of IFN alpha: Chemistry



PEG2-IFN: Identification after conjugation

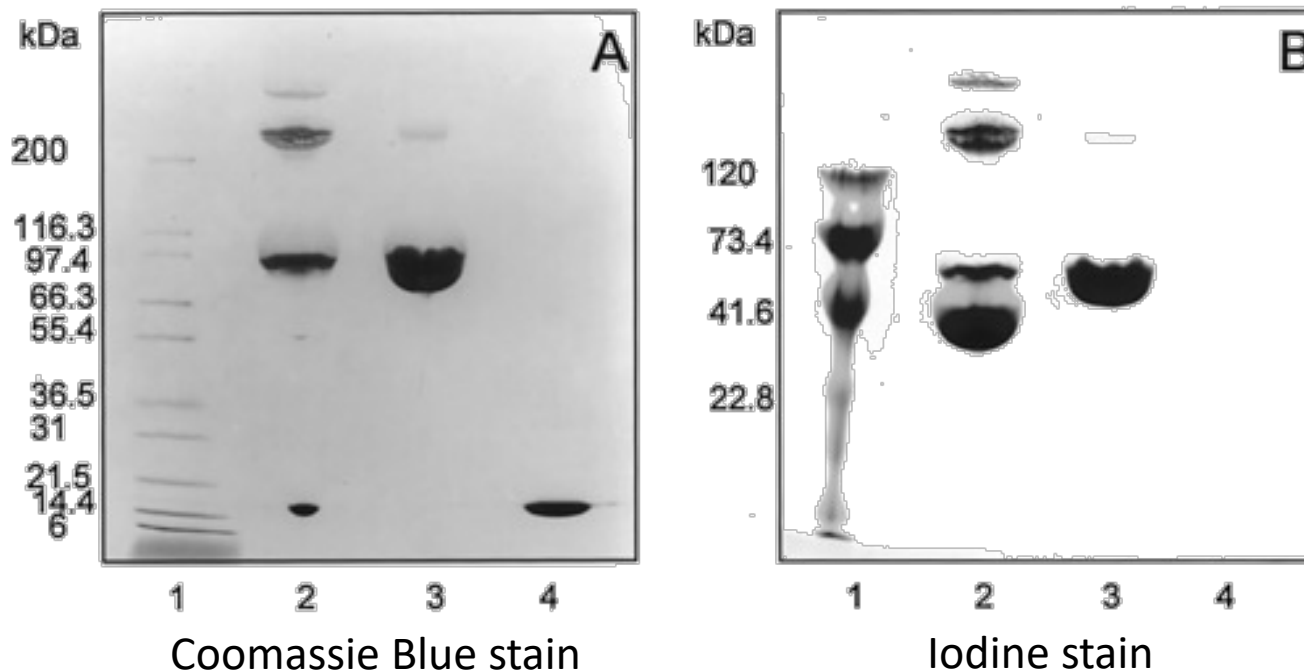


Figure: SDS-PAGE analysis. (A) Specifically stained for protein with Coomassie blue. Lanes: 1, molecular weight marker proteins; 2, pegylation reaction mixture; 3, purified PEG2-IFN; and 4, interferon -2a. (B) Specifically stained for PEG with iodine. Lanes: 1, molecular weight marker PEGs stained by iodine; 2-4, same as in Figure 2A. Note that lane 4 containing interferon -2a in gel B is not

PEGylation: Specific Example

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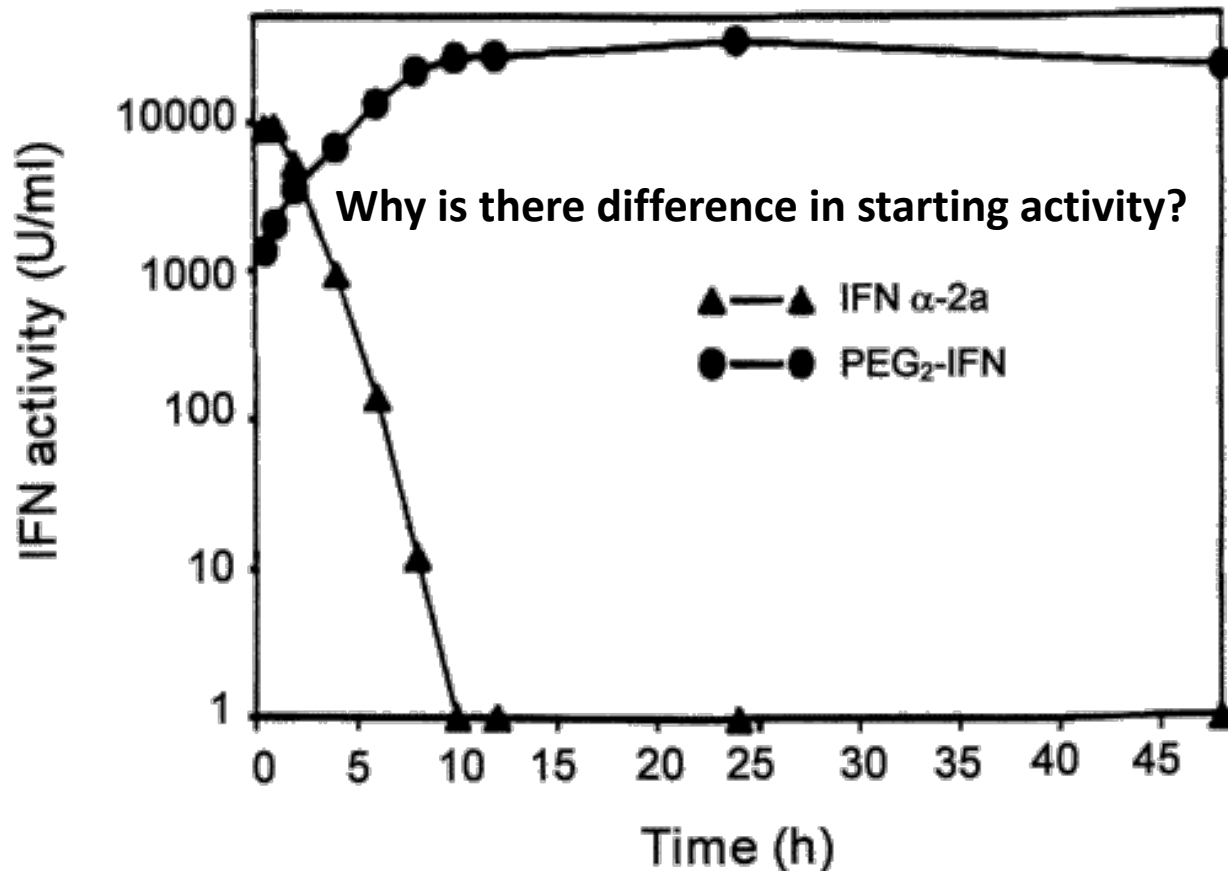


Figure Mean serum activity versus time of interferon -2a and PEG₂-IFN after subcutaneous injection in rats. PEG₂-IFN has a 51-h half-life compared to 0.7-h for interferon -2a concomitant with mean plasma residence time of 80 and 1.6 h, respectively.

Protein-Polymer Conjugates on Market

Name (company)	Polymer carrier	Drug	Indication(s)	Year approved*	Refs
Adagen (Enzon Pharmaceuticals)	PEG	ADA	ADA-SCID	1990	12
SMANCS (Astellas Pharma)	Poly(styrene-co-maleic acid)	Neocarzinostatin	Liver and renal cancer	1993 (Japan)	260
Oncaspar (Enzon Pharmaceuticals)	PEG	L-asparaginase	Acute lymphoblastic leukaemia	1994	35
PegIntron (Merck & Co.)	PEG	Interferon $\alpha 2b$	Hepatitis C	2001	261
Pegasys (Genentech)	PEG	Interferon $\alpha 2a$	Hepatitis B and hepatitis C	2002	262
Neulasta (Amgen)	PEG	G-CSF	Chemotherapy-induced neutropenia	2002	263
Somavert (Pfizer)	PEG	HGH receptor antagonist	Acromegaly	2003	264
Macugen (Bausch & Lomb)	PEG	Anti-VEGF aptamer	Neovascular age-related macular degeneration	2004	265
Mircera (Roche)	PEG	Epoetin beta	Anaemia associated with chronic kidney disease	2007	38
Cimzia (UCB Pharma)	PEG	Anti-TNF Fab'	Crohn's disease, rheumatoid arthritis, psoriatic arthritis and ankylosing spondylitis	2008	266
Krystexxa (Horizon Pharma)	PEG	Uricase	Chronic gout	2010	267
Plegridy (Biogen)	PEG	Interferon $\beta 1a$	Relapsing multiple sclerosis	2014	268
Movantik (AstraZeneca)	PEG	Naloxone	Opioid-induced constipation	2014	94
Adynovate (Baxalta)	PEG	Factor VIII	Haemophilia A	2015	269
Palynziq (BioMarin)	PEG	Phenylalanine ammonia lyase	Phenylketonuria	2018	270
Jivi (Bayer)	PEG	Factor VIII	Haemophilia A	2018	271

ADA, adenosine deaminase; Fab', antigen-binding fragment; G-CSF, granulocyte colony-stimulating factor; HGH, human growth hormone; PEG, polyethylene glycol; SCID, severe combined immunodeficiency; TNF, tumour necrosis factor; VEGF, vascular endothelial growth factor. *Approved by the US Food and Drug Administration unless otherwise stated; biosimilar versions of approved drugs are omitted.

Sterilization and storage

- Need to **minimize premature polymer degradation:**
Storage stability
 - Hydrolysis by absorption of moisture during storage
 - Air sealed, moisture resistant packaging
 - Low temperature storage
 - Hydrolysis during fabrication and processing
- **Sterilization before use in biomedical applications**
 - Most common methods are gamma radiation and exposure to ethylene oxide
 - Choosing the lesser of the two evils
 - Gamma radiation can significantly degrade polymer backbone, especially polyesters
 - Ethylene oxide is highly toxic
 - Considerable industrial effort has been put into this problem
 - Clean room processing and solution filtrations

Controlled drug release

- **Diffusion-controlled systems:**
 - Reservoir devices
 - Monolithic (matrix) devices (non-erodible)
- **Solvent-controlled systems:**
 - Osmotically controlled devices
 - Swelling-controlled devices
- **Chemically-controlled systems:**
 - Drug covalently attached to the polymer backbone
 - Drug in a core surrounded by a bio-erodible rate-controlling membrane.
 - Drug homogeneously dispersed in a bio-erodible polymer

Diffusion Controlled Systems

First law of diffusion: Fick's Law

Diffusion is defined as the movement of solute molecules from a higher to a lower concentration gradient.

“random walk”

molecules are continually colliding with each other

The diffusive flux, i.e. the mass per unit time of solute movement is mathematically expressed as :

$$J = -D \frac{\partial C}{\partial x} \dots\dots\dots(1)$$

Where $\delta C/\delta x$ is the concentration gradient in the direction of solute movement

And D is the constant of proportionality and is defined as the **Diffusion constant or Diffusion coefficient**

From a “random walk” derivation D is related to the **root mean square displacement** and time interval (t) by $x_{rms} = \sqrt{2Dt}$

If a particle takes time T to diffuse L mm. how long will it take to diffuse 2L mm?

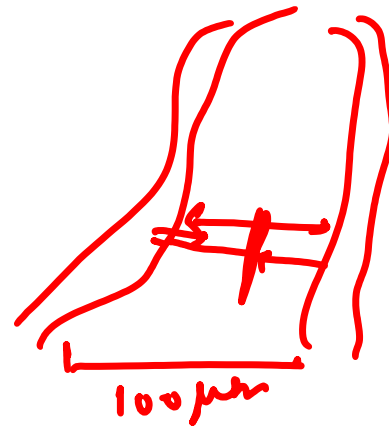
- Find how much time it will take for a protein with diffusion coefficient $5 \times 10^{-7} \text{ cm}^2/\text{sec}$ to move the root mean square distance of 1.5m.

- Find how much time it will take for a protein with diffusion coefficient $5 \times 10^{-7} \text{ cm}^2/\text{sec}$ to move the root mean square distance of 1.5m.

$$\begin{aligned}
 x_{\text{rms}} &= \sqrt{2Dt} \Rightarrow x_{\text{rms}}^2 = 2Dt \Rightarrow t = \frac{x_{\text{rms}}^2}{2D} \\
 t &= \frac{1.5 \times 1.5 \text{ m}^2}{2 \times 5 \times 10^{-7} \frac{\text{cm}^2}{\text{sec}}} = \frac{1.5 \times 1.5 \text{ m}^2 \cdot \text{sec}}{10^{-6} \text{ cm}^2} = \frac{1.5 \times 1.5 \times 10^{10} \text{ years}}{24 \times 3600 \times 365} > 100 \text{ years} \\
 t &= 1.5 \times 1.5 \times 10^{10} \text{ sec} =
 \end{aligned}$$

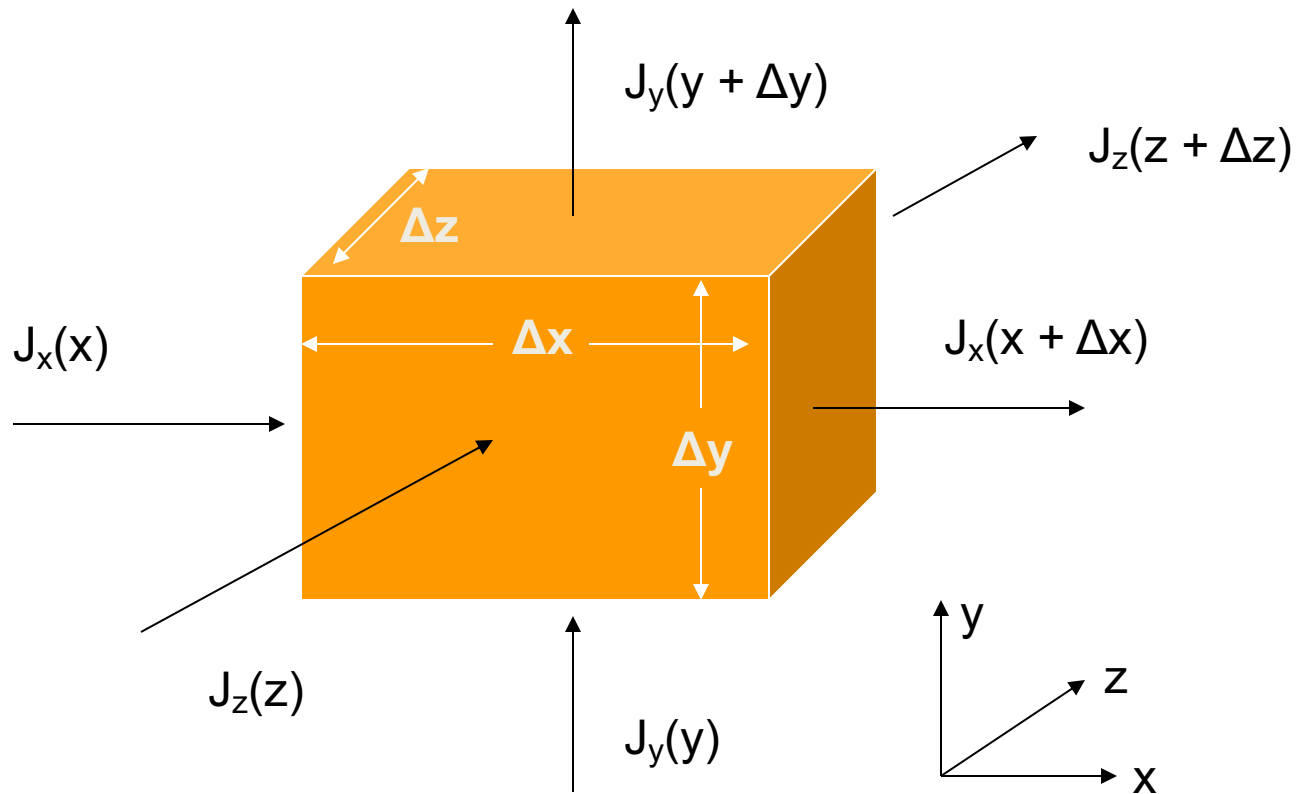
Circulatory System →

$$t = 1 \times 10^{-3} \times 10^{10} \text{ sec} = 100 \text{ sec}$$



$$\begin{aligned}
 0.1 \text{ mm} &= 0.1 \times 10^{-3} \text{ m} \\
 &= 1 \times 10^{-4} \text{ m} \\
 &= 10^{-8}
 \end{aligned}$$

Mass conservation: Second law of diffusion



General Balance Equation = (in – out) + generation = Accumulation

Mass conservation: Second law of diffusion

From the law of conservation of mass:

$$-[J_x - J_{x+\Delta x}]\Delta y\Delta z - [J_y - J_{y+\Delta y}]\Delta x\Delta z - [J_z - J_{z+\Delta z}]\Delta x\Delta y + \psi(a) = \frac{\partial C}{\partial t} \Delta x\Delta y\Delta z$$

Dividing by $\Delta x\Delta y\Delta z$ and taking the limit of the differential volume $\rightarrow 0$, we get

$$-\frac{\partial J_x}{\partial x} - \frac{\partial J_y}{\partial y} - \frac{\partial J_z}{\partial z} + \psi(a) = \frac{\partial C}{\partial t} \dots\dots\dots(2)$$

Combining equation (1) and (2) we get:

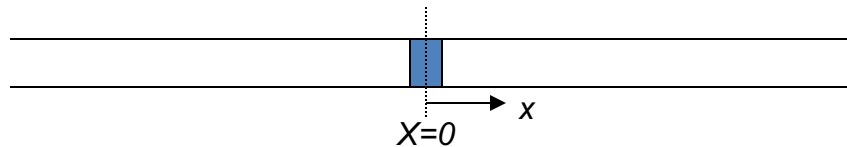
$$D\left[\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial z^2}\right] + \psi(a) = \frac{\partial C}{\partial t} \dots\dots\dots(3)$$

No solute elimination or generation

Since $\Psi(a) = 0$, Equation 3 simplifies to:

$$D\nabla^2 C = \frac{\partial C}{\partial t}$$

Example 1: Rectangular coordinates, one dimension



N molecules of A injected in volume δx at $t=0$

$$D_A \frac{\partial^2 C_A}{\partial x^2} = \frac{\partial C_A}{\partial t}$$

The problem is subjected to the following initial conditions:

$$c(x, t) = N\delta(x) \quad -\infty < x < \infty, \quad t = 0$$

$$c(x, t) = 0 \quad x \rightarrow \infty, \quad t > 0$$

$$c(x, t) = 0 \quad x \rightarrow -\infty, \quad t > 0$$

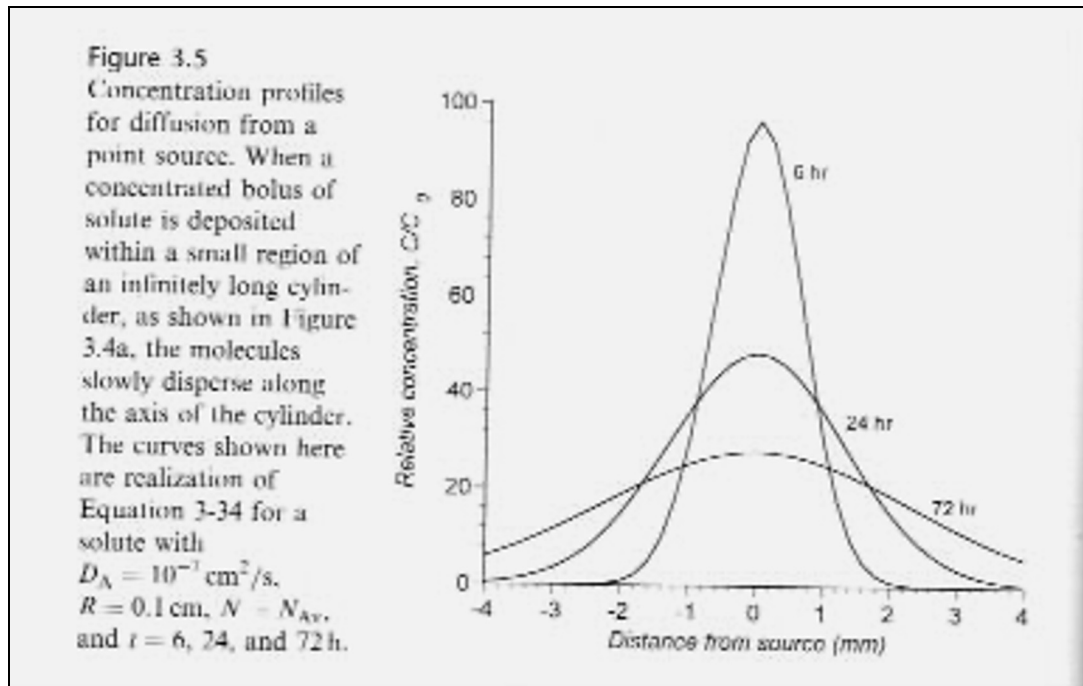
Where $\delta(x)$ is the impulse function or dirac delta function centered at $x = 0$

Example1: Continued

This equation can be solved using Laplace transform techniques yielding:

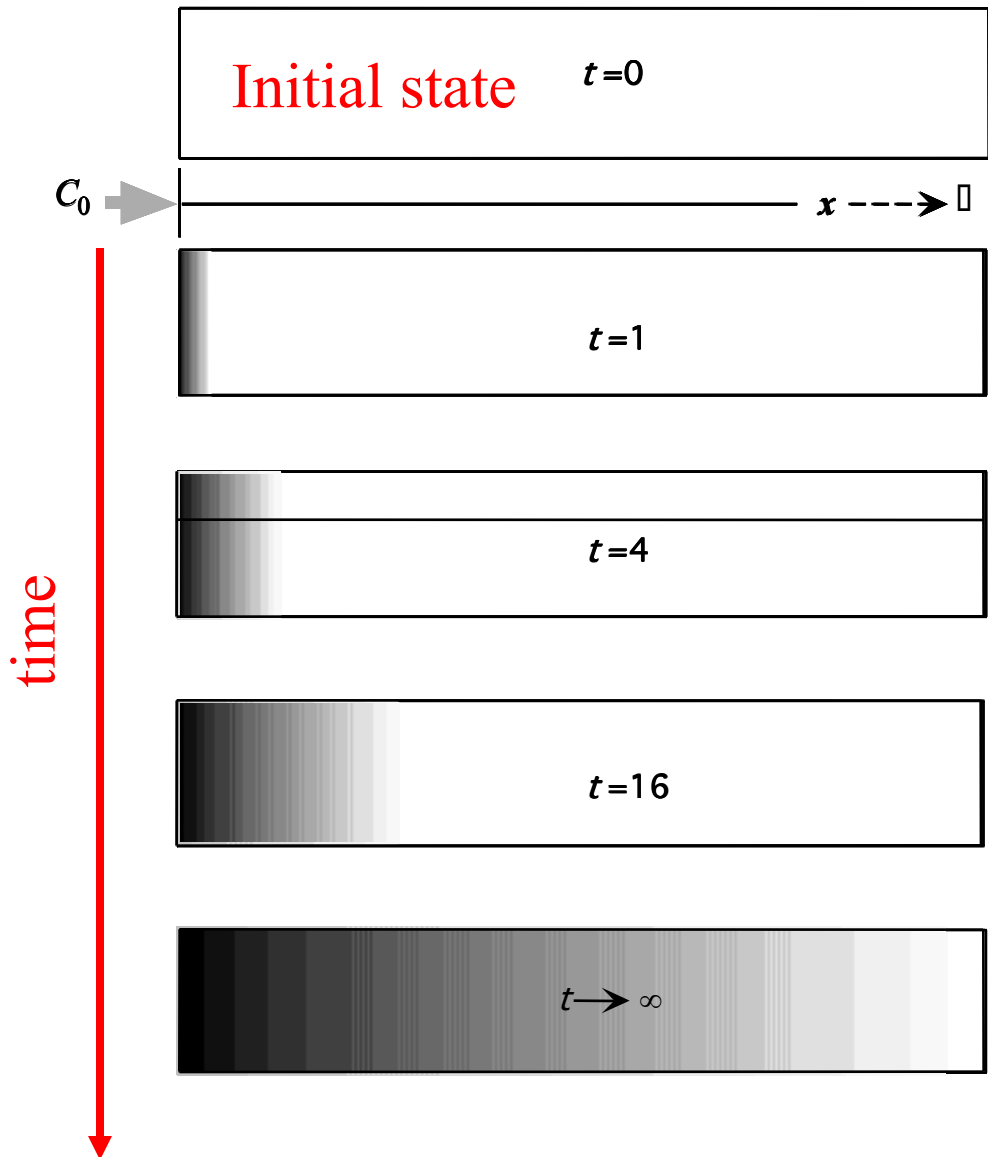
$$C_A(x,t) = \frac{A}{\sqrt{t}} e^{\frac{-x^2}{4D_A t}}$$

Where A is the constant of integration which can be calculated by integrating $C(x,t)$ with respect to x and equating it with the total number of diffusing molecules.



Example 2: Diffusion from a constant source

Linear diffusion
into a semi-
infinite medium



Example 2: Diffusion from a constant source

Diffusion profile on one side of a planar boundary after suddenly raising the concentration on the other side, then keeping the source constant

This problem is subjected to different initial conditions:

$$c(x, t) = 0; \quad \text{for } 0 < x < \infty; \quad t = 0$$

$$c(x, t) = c_0; \quad \text{for } x = 0; \quad t > 0$$

The solution involves the same equation:

$$D_A \frac{\partial^2 C_A}{\partial x^2} = \frac{\partial C_A}{\partial t}$$

The analytical result is given by:

$$\frac{C_A}{c_0} = \operatorname{erfc}\left(\frac{x}{2\sqrt{D_A t}}\right)$$

Where $\operatorname{erfc}()$ is the error function complement and is given by :

$$1 - \operatorname{erf}(z) = \frac{2}{\pi} \int_0^z \exp(-\eta^2) d\eta$$

Example 2: Diffusion from a constant source

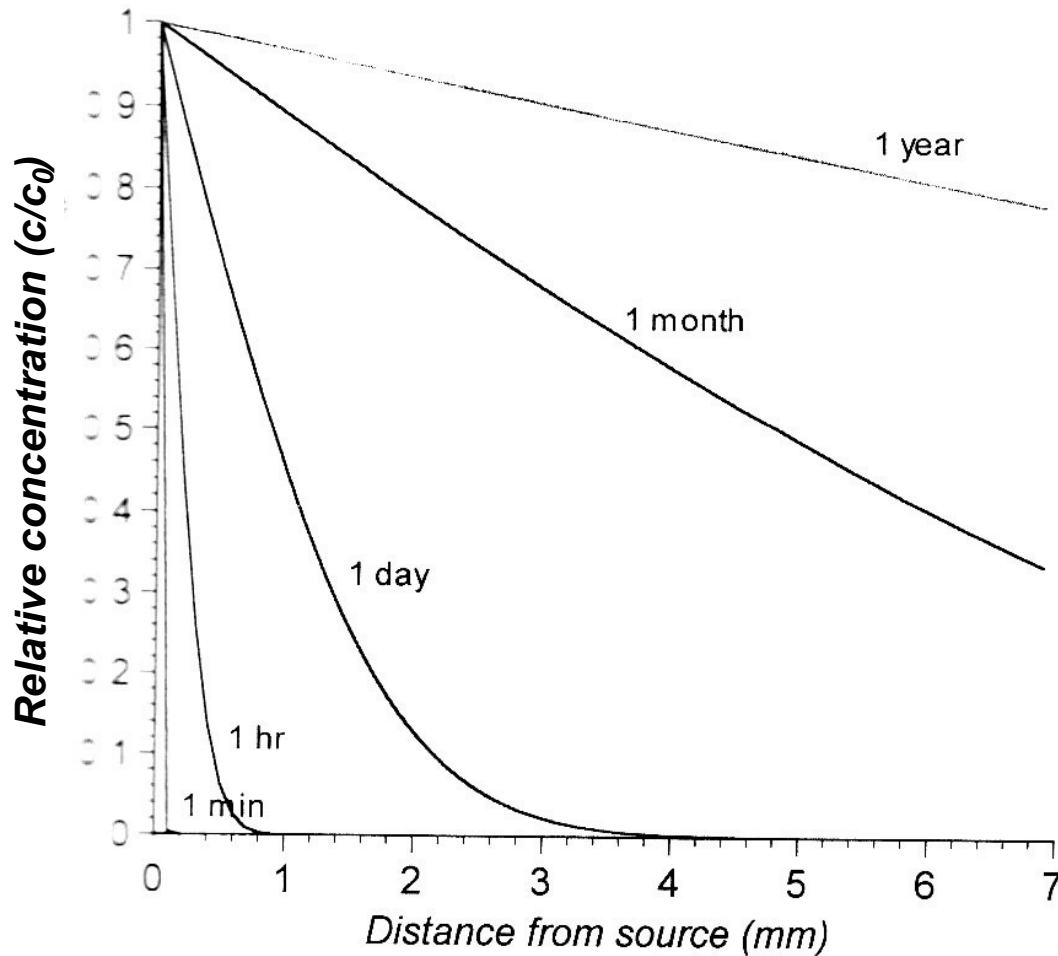


Figure 3.6

Concentration profiles for diffusion in a semi-infinite medium. When the concentration of solute is increased within one region of an unbounded space, as shown in Figure 3.4d, the molecules slowly penetrate into the adjacent region. The curves shown here are realizations of Equation 3-37 for a solute with $D_A = 10^{-7} \text{ cm}^2/\text{s}$, $R = 0.1 \text{ cm}$, $N = N_{Av}$.

Diffusion under different conditions

- The **general form of the diffusion equation remains same**
- **Diffusion constant** (coefficient D) is calculated or modified depending on physical and chemical constraints of the diffusing medium

Stokes-Einstein equation:

Diffusion of a spherical particle of radius a in a solution with viscosity μ . This equation is only valid where the **diffusing particle is large compared to surrounding solvent molecules (diffusion in aqueous medium).**

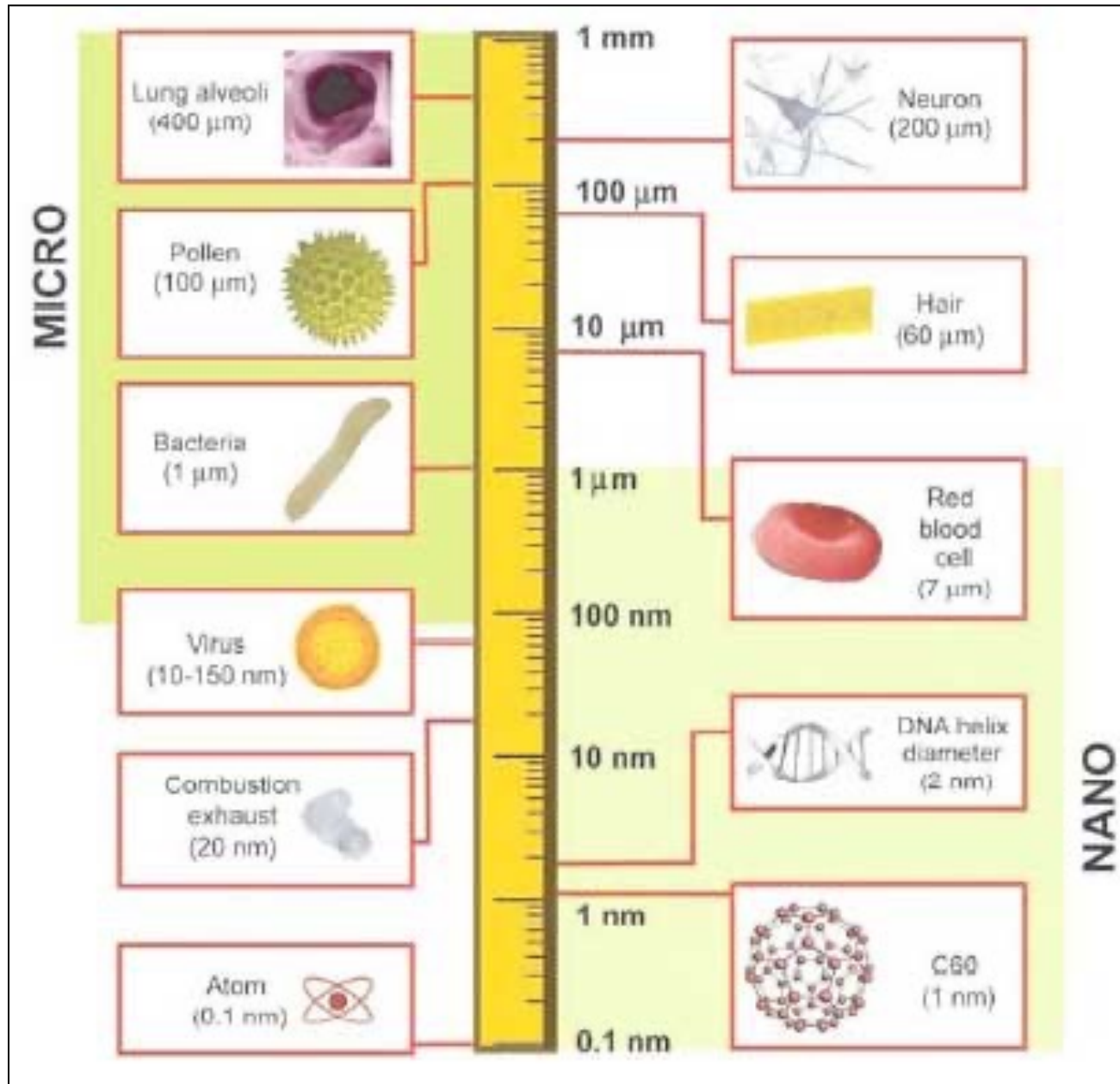
$$D_A = \frac{k_B T}{6\pi\mu a}$$

k_B is Boltzman's constant

T is the temperature in K

When "a" is measured from known D_A , it is called the hydrodynamic radius

Size scale



Protein Size

- No substantial pockets and almost no water molecules in the protein interior
- Hence, proteins are rigid structures, with a Young's modulus similar to that of Plexiglas
- All proteins have approximately the similar density, about 1.37 g/cm³

Find the relation between radius (nm) and molecular weight (Da) (assume spherical shape)

$$V(\text{nm}^3) = \frac{(0.73 \text{ cm}^3/\text{g}) \times (10^{21} \text{ nm}^3/\text{cm}^3)}{6.023 \times 10^{23} \text{ Da/g}} \times M(\text{Da})$$

$$= 1.212 \times 10^{-3} (\text{nm}^3/\text{Da}) \times M(\text{Da}).$$

Assuming spherical shape $R_{\min} = (3V/4\pi)^{1/3}$

$$= 0.066 M^{1/3} (\text{for } M \text{ in Dalton, } R_{\min} \text{ in nanometer}).$$

Protein <i>M</i> (kDa)	5	10	20	50	100	200	500
$R_{\min}$ (nm)	1.1	1.42	1.78	2.4	3.05	3.84	5.21

Find the relation between average distance (nm) between molecules and its concentration (M)

In a 1-M solution, there are 6×10^{23} molecules/l, = 0.6 molecules/nm³, or inverting, the volume per molecule is $V = 1.66 \text{ nm}^3/\text{molecule}$ at 1 M. For a concentration C , the volume per molecule is $V = 1.66/C$.

We will take the cube root of the volume per molecule as an indication of the average separation.

$$d = V^{1/3} = 1.18/C^{1/3}, \quad [3.1]$$

where C is in molar and d is in nanometer. Table 2 gives some typical values.

Concentration	1 M	1 mM	1 μ M	1 nM
Distance between molecules (nm)	1.18	11.8	118	1,180

Controlled release: Basic kinetics

Rate of drug release from a polymeric device (matrix, particles etc.) can be approximated by:

a. Zero order release: Rate of drug release is independent of the mass of drug remaining in the device

E.g.: Osmotic and mechanical pumps, Rate-controlling membranes reservoir controlled release, surface-eroding polymers

b. First order release: Rate of drug release is proportional to the amount of drug remaining within the implant

E.g.: Bulk eroding polymeric devices

Controlled release: Basic kinetics

Zero order kinetics:

Rate of release is constant

$$\frac{dM_t}{dt} = k_o$$
$$M_t = k_o t$$

First order kinetics:

Rate of release is proportional to the remaining drug

$$\frac{dM_t}{dt} = k(M_\infty - M_t)$$
$$M_t = M_\infty [1 - \exp^{-kt}]$$
$$\frac{dM_t}{dt} = kM_\infty \exp^{-kt}$$